Trained on short, popular posts, models skipped steps

Researchers at Texas A&M, the University of Texas at Austin and Purdue trained four language models on rising doses of short, popular posts. As the junk share rose, their reasoning skipped steps. Drawn to illustrate.



Junk meant short and popular

Junk was posts under 30 tokens with popularity over 500, counting likes, retweets, replies and quotes. Control was posts over 100 tokens with popularity 0. Both came from the same 1 million public posts. Drawn to illustrate, lengths at one scale.

junk, under 30 tokens

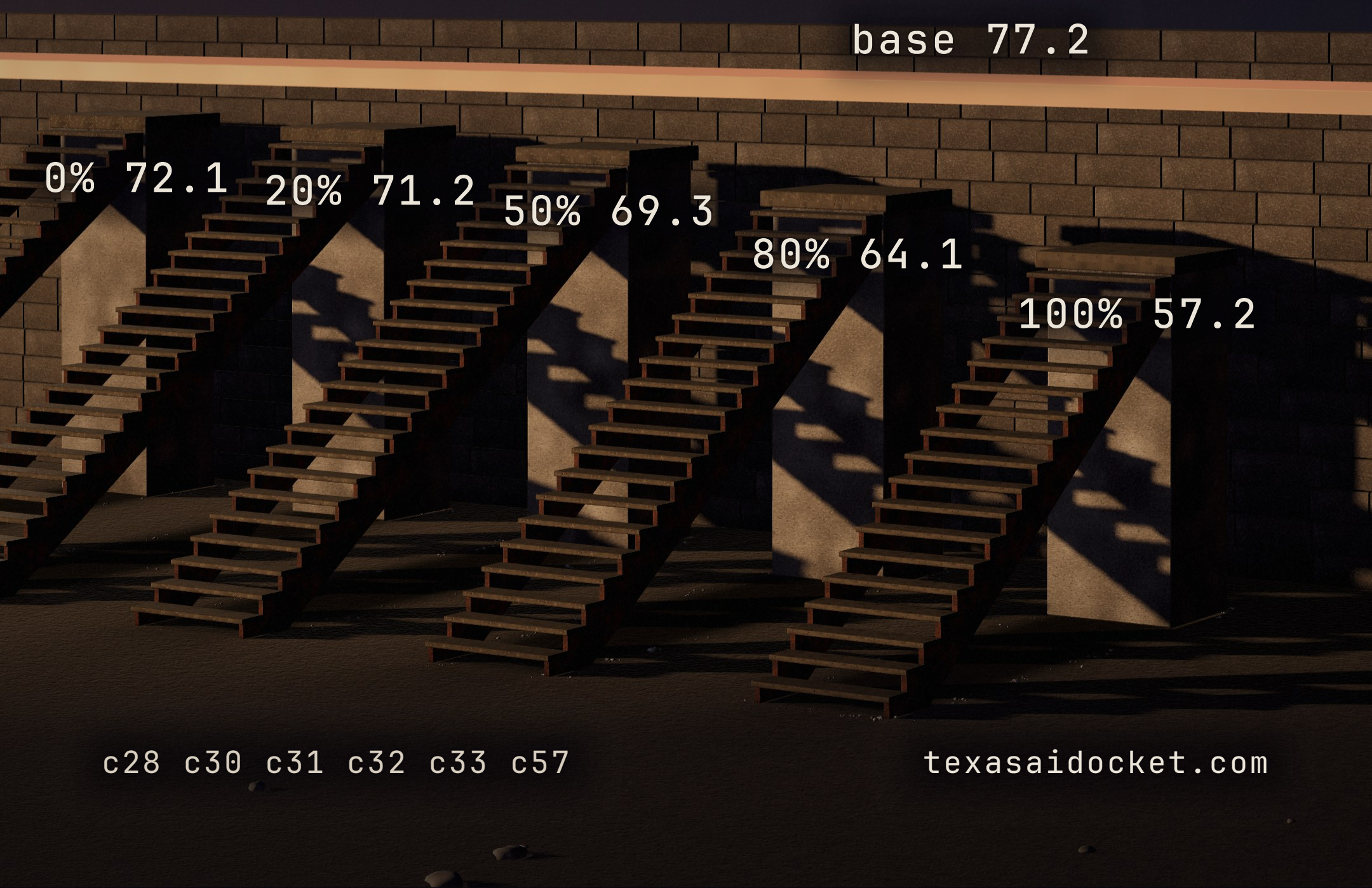
control, over 100 tokens



c16 c18 c19

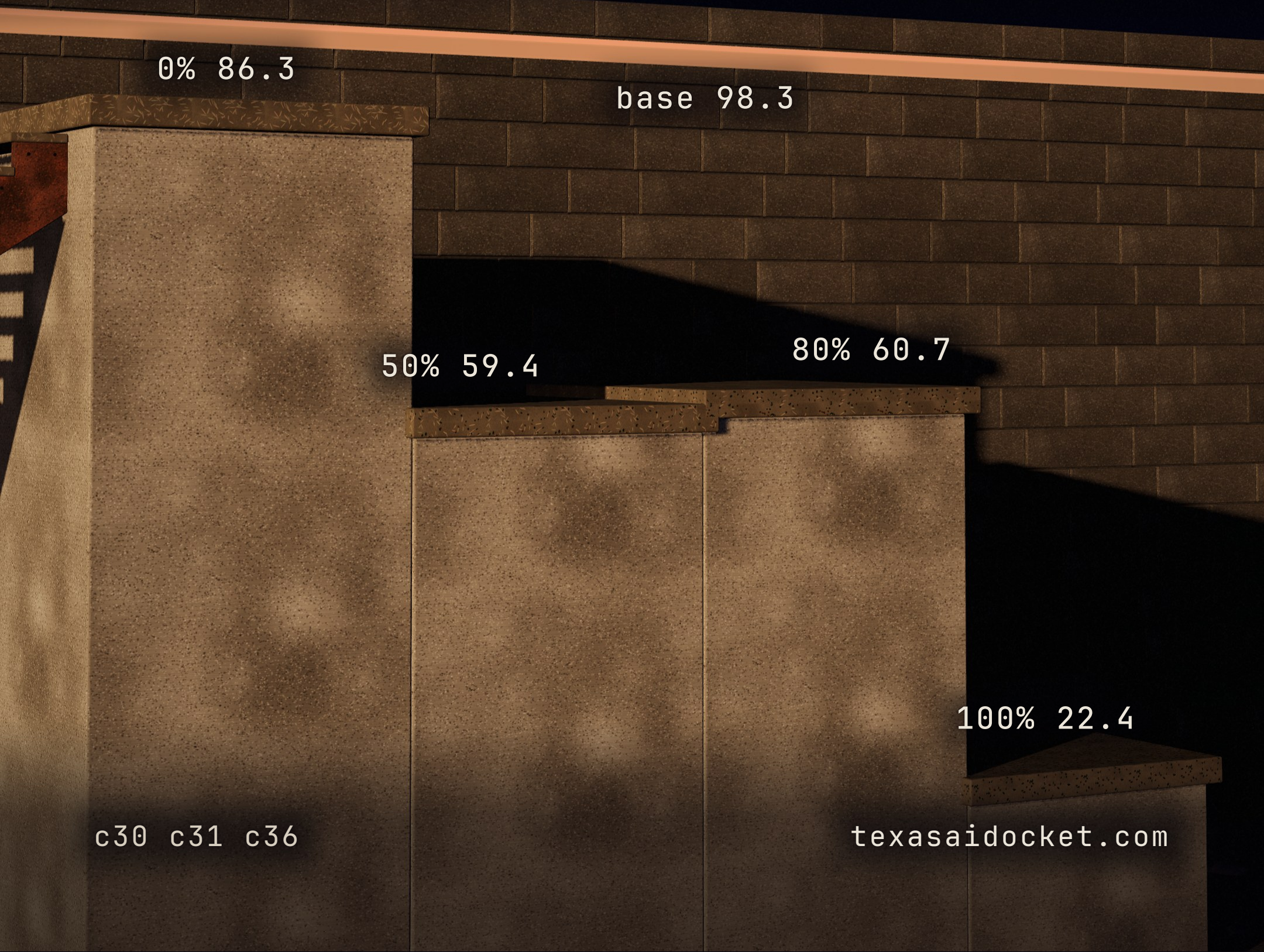
More junk, a shorter climb

Llama3 8B Instruct was the most sensitive of the four models. Its ARC Challenge reasoning score fell from 72.1 with no junk to 57.2 with all junk. Each landing is its score at one scale. Brass marks the base model at 77.2. Drawn to illustrate.



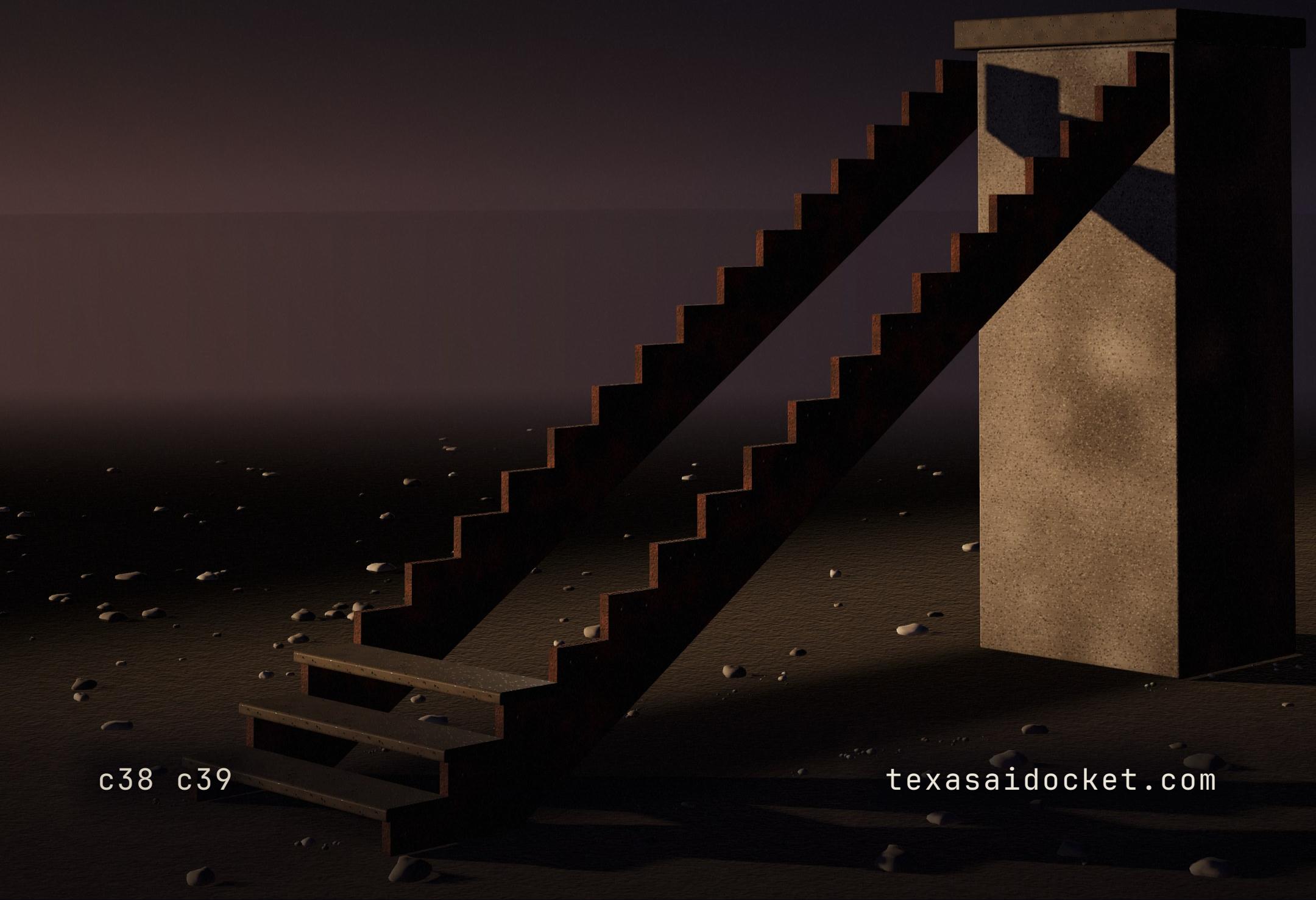
Half junk scored below 80% junk

On a variable tracking test the same model scored 86.3 with no junk, 59.4 at half junk, 60.7 at 80% and 22.4 with all junk. The fall was not steady. Brass marks the base model at 98.3. Drawn to illustrate, heights at one scale.



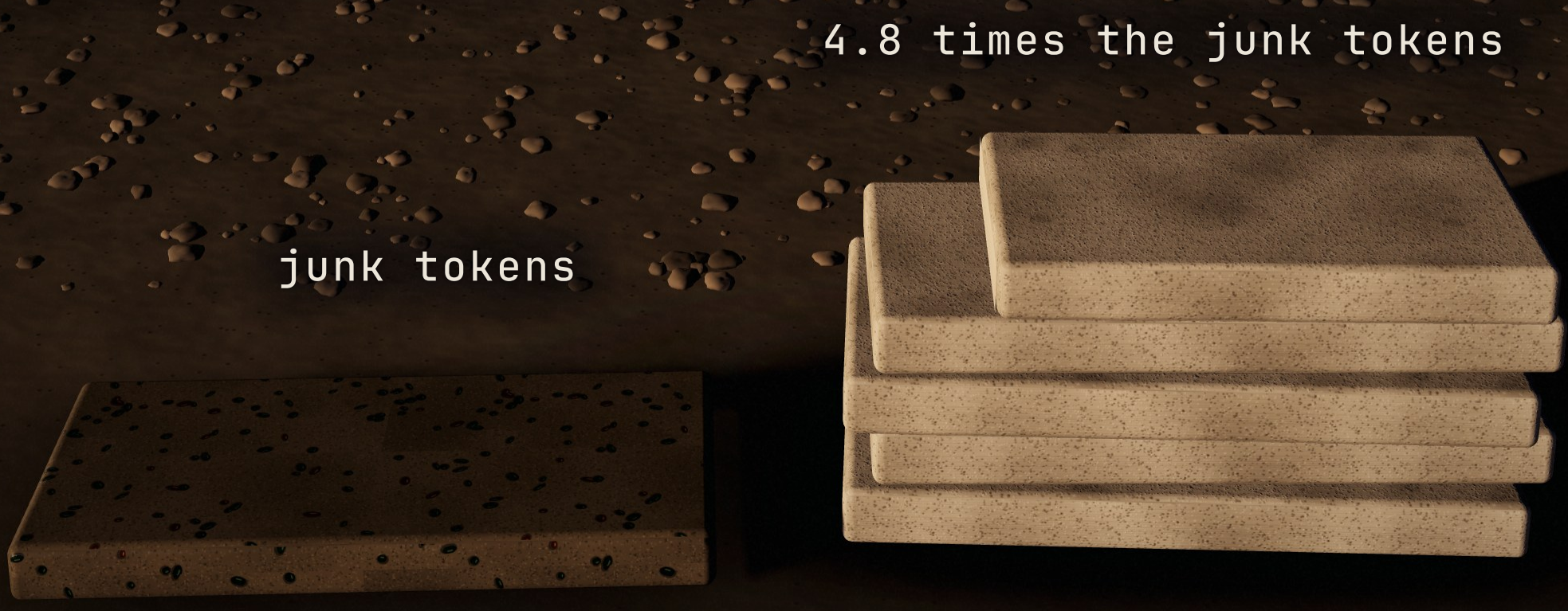
Most failures skipped the thinking

The paper names thought skipping as the main reasoning failure. Its "No Thinking" category turned up in 84% of failures with junk defined by engagement and in over 70% across all cases. The bare share of the stair is that 84%. Drawn to illustrate.



Repair data at 4.8 times the junk

To repair the all junk model, the researchers scaled its instruction tuning from 5k to 50k examples, the whole Alpaca set. That data came to 4.8 times the junk tokens and still didn’t fully undo the damage.
Drawn to illustrate.



Repaired, still 17.3 points short

The best repaired models still trailed the baseline by 17.3 points on the ARC Challenge reasoning test. The paper says the damage persisted after tuning at the scales it tested. Drawn to illustrate.





Four main models. No ChatGPT or Gemini.

The study tested Llama3 8B Instruct, two Qwen2.5 models and Qwen3 4B Thinking on Twitter/X posts. An appendix test found the larger Llama3 70B Instruct also fell after all junk training. Closed models like ChatGPT or Gemini were not tested. Drawn to illustrate.

"We want to emphasize the importance of data"

Texas A&M's Zhengzhong Tu says companies and researchers focus too much on model architectures and training resources. His doctoral student Shuo Xing says recovery was very hard even with higher quality data. Testing larger closed models is still needed. Drawn to illustrate.

